

BARSHATAS, Kazakhstan

WMO: 364980

Lat: 48.162N

Lon: 78.675E

Elev: 2110

StdP: 13.61

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-14.6	-24.5	1.5	-18.0	-20.4	1.9	-13.6	27.9	7.1	23.5	13.1	5.2	50	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	90.1	61.9	87.3	61.1	84.4	60.2	64.6	82.3	63.1	81.1	61.9	79.7	6.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	80.5	68.5	57.0	74.8	69.0	54.7	68.8	69.1	30.8	82.5	29.7	81.2	28.7	79.3	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	20.5	16.2	DB	-26.3	96.5	7.9	2.2	-32.0	98.1	-36.6	99.4	-41.1	100.6	-46.9	102.2	(4)
(5)			WB	-24.7	68.3	5.3	2.1	-28.5	69.8	-31.7	71.1	-34.7	72.2	-38.5	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	6.8	7.1	21.1	43.3	55.5	66.5	70.5	66.6	55.7	41.4	24.3	11.1
	DBStd	24.92	9.80	9.93	11.22	9.56	8.79	7.38	6.03	6.51	8.20	9.20	11.31	11.11
	HDD50	6034	1338	1201	896	239	49	2	0	1	38	294	770	1206
	HDD65	9810	1803	1621	1360	652	310	69	19	59	293	733	1220	1671
	CDD50	2141	0	0	1	38	219	496	637	515	208	27	0	0
	CDD65	440	0	0	0	0	15	114	190	108	13	0	0	0
	CDH74	5257	0	0	0	11	238	1357	2104	1288	250	9	0	0
	CDH80	1785	0	0	0	1	46	455	787	443	53	0	0	0
Wind	WSAvg	4.7	4.0	4.6	4.7	5.5	5.3	5.3	5.0	5.1	4.8	4.3	3.9	3.9
Precipitation	PrecAvg	8.1	0.6	0.6	0.4	0.6	0.7	0.7	1.5	0.6	0.2	0.4	0.8	0.7
	PrecMax	12.4	1.2	1.3	1.1	1.5	1.5	1.5	4.5	1.8	0.8	1.3	1.9	1.4
	PrecMin	5.5	0.1	0.2	0.1	0.0	0.1	0.0	0.2	0.0	0.0	0.1	0.2	
	PrecStd	1.9	0.3	0.3	0.3	0.4	0.4	0.4	1.2	0.6	0.2	0.4	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	33.1	58.5	75.8	85.1	92.0	96.3	92.9	86.1	73.3	51.5	34.8
		MCWB	27.1	30.6	47.2	54.1	58.7	63.3	64.7	62.1	58.8	53.1	41.2	31.7
	2%	DB	25.8	29.0	47.6	68.8	80.0	88.4	91.0	88.8	80.3	66.9	46.7	32.1
		MCWB	23.6	26.3	39.8	51.3	57.7	61.6	62.2	60.7	56.4	49.4	38.8	30.4
	5%	DB	23.2	25.8	41.3	64.5	76.1	85.5	87.6	85.2	76.5	62.3	42.2	29.6
		MCWB	21.1	23.3	35.7	49.2	55.9	60.3	61.7	59.7	54.8	47.4	36.7	27.8
	10%	DB	20.7	22.5	36.4	60.0	72.2	82.0	84.6	81.5	72.6	57.1	37.9	26.5
		MCWB	18.9	20.3	32.1	47.0	53.7	59.0	61.3	58.9	53.2	45.0	33.4	24.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	31.4	47.8	56.6	61.7	65.7	67.3	64.6	60.7	54.0	43.5	32.8
		MCDB	29.8	32.9	57.4	71.8	79.5	88.2	86.1	82.7	84.5	72.0	49.1	33.8
	2%	WB	23.9	26.6	40.5	52.4	59.1	63.7	65.3	62.9	57.2	50.3	39.9	30.5
		MCDB	25.7	28.9	47.0	66.8	75.6	82.9	82.2	81.5	77.1	65.3	44.8	32.1
	5%	WB	21.3	23.5	36.2	50.0	57.1	62.1	63.8	61.4	55.4	47.7	36.8	27.9
		MCDB	23.1	25.8	41.0	62.9	72.8	80.7	81.0	79.9	74.8	60.7	42.2	29.7
	10%	WB	19.0	20.3	32.2	47.5	55.2	60.5	62.5	60.0	53.6	45.4	33.6	24.6
		MCDB	20.8	22.4	35.9	58.9	69.7	77.9	80.0	78.4	71.1	56.7	37.7	26.4
Mean Daily Temperature Range	5% DB	MDBR	15.4	17.7	17.4	20.8	22.7	22.7	22.3	23.0	23.4	19.7	16.5	14.9
		MCDBR	15.8	17.9	19.9	27.5	27.6	26.2	26.4	28.5	29.1	27.4	21.4	15.0
		MCWBR	14.4	15.9	13.7	15.0	13.2	10.3	9.4	10.6	13.3	15.1	14.6	13.1
	5% WB	MCDBR	15.5	17.2	19.5	25.2	24.7	23.4	23.0	24.6	27.5	25.8	19.3	13.3
		MCWBR	14.2	15.7	13.6	14.9	14.2	10.8	9.5	10.7	13.3	15.1	14.0	12.1
Clear-Sky Solar Irradiance	taub	0.286	0.316	0.370	0.413	0.411	0.415	0.430	0.387	0.344	0.342	0.322	0.289	
	taud	2.102	2.047	2.071	2.099	2.159	2.217	2.188	2.294	2.370	2.285	2.198	2.130	
	Ebn at Noon	240	259	263	264	270	269	263	270	272	250	225	221	
	Edh at Noon	28	37	43	47	46	43	44	38	32	29	25	24	
All-Sky Solar Radiation	RadAvg	565	920	1340	1662	2109	2160	2033	1909	1472	879	492	402	
	RadStd	36	49	86	108	103	127	126	85	96	74	43	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page